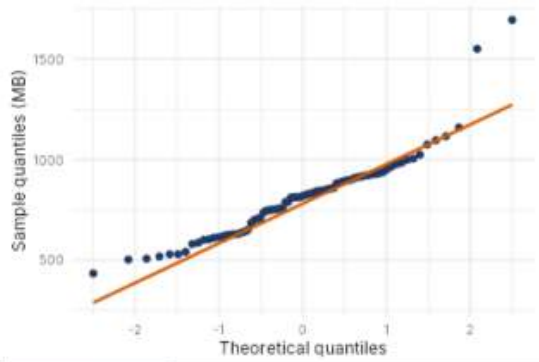
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot C

R.No: 6 Reg. No: 192325016 Name: BANDLAPALLI BHANUTEJA REDDY

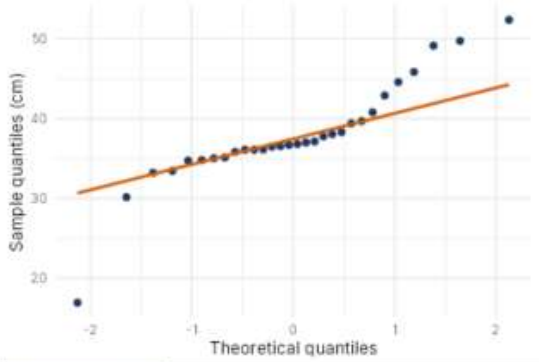
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: mobile data used per day (MB), n = 80



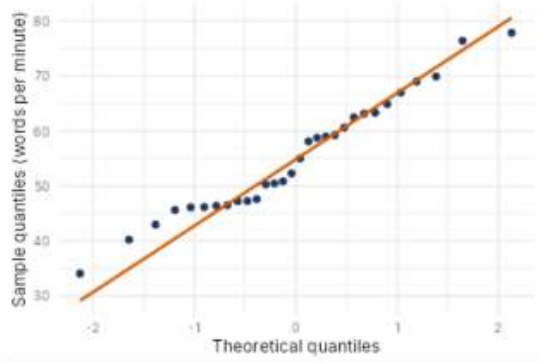
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: plant height (cm), n = 30



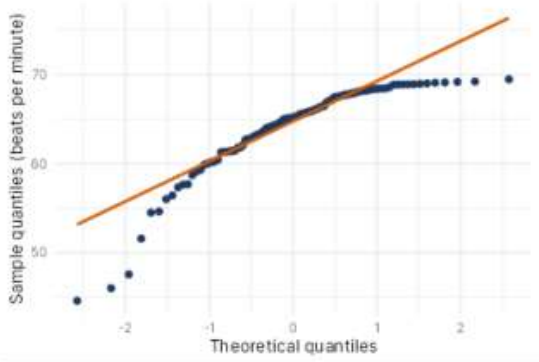
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: typing speed (words per minute), n = 30



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: resting pulse (beats per minute), n = 100



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

LAB ACTIVITY RECORD

Activity 5: Normality Detection using Q-Q Plots

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Activity 5 - Normality Detection using Q-Q Plots
Name	BANDLAPALLI BHANUTEJA REDDY
Register Number	192325016
Date	13 OCTOBER 2026
Day	Tuesday

Aim

To look at four normal Q-Q plots made in R, to decide for every chart whether the data are normal, to name the shape when they are not (and give the evidence seen in the plot), and to answer two short questions on the remedy for non-normal data and on the ggplot2 code for a Q-Q plot.

What Is a Normal Q-Q Plot?

A Q-Q plot (quantile-quantile plot) compares my data with a perfect normal distribution. Each dot is one data value. The x-axis (theoretical quantiles) is where that value would sit if the data were exactly normal, and the y-axis (sample quantiles) is where it really sits. The red line is the "perfect normal fit" (drawn by stat_qq_line, through the first and third quartiles). So the rule is very simple: if the dots lie along the line, the data are normal; the way the dots leave the line tells me the shape.

How I Read the Shapes

What the dots do	Left end	Right end	Shape
Follow the line closely	On the line	On the line	Normal
Bend upward like a smile (curve above the line at both ends)	Above the line	Above the line, far away	Right-skewed (long tail to the right)
Bend downward like a frown (below the line at both ends)	Below the line, far away	Below the line	Left-skewed (long tail to the left)
S-shape: left end below, right end above the line	Below the line	Above the line	Heavy-tailed (more extreme values than normal)
Flat S-shape: left end above, right end below the line	Above the line	Below the line	Light-tailed (fewer extreme values)
Two separate groups of dots with a jump between them	Steps or a gap in the middle		Bimodal
Most dots on the line, only one or two far away	On the line	Few single dots away	Normal with outlier(s)

The question sheet carries 16 marks for Part A (4 marks per chart: normal or not, shape, and the evidence) and 4 marks for Part B. Below, every chart is answered in the same three boxes that the sheet gives, followed by a short explanation of how I reached the answer.

Chart 1: Mobile Data Used per Day (MB), n = 80

Aim: To decide whether the daily mobile data use of 80 users is normal, and to name the shape.

What I See in the Plot

(a) The dots in the middle (about 600 to 1000 MB) lie on or very close to the red line.

(b) At the left end the dots (about 430 to 650 MB) sit above the line, so the small values are not as small as a normal curve would give.

(c) At the right end the dots climb steeply above the line. The two largest values (about 1550 MB and 1700 MB) are far above the line, which would pass near 1200 to 1270 MB there.

Answer (as asked in the sheet)

Normal? (Yes / No)	**No**
Shape (from the list)	Right-skewed
Evidence in the plot	The dots curve upward above the line at both ends, and the right end shoots far above it (two large users at about 1550 and 1700 MB). The left end is squeezed above the line, and the middle follows the line.

Explanation

Step 1: Look at the middle of the plot first. The dots follow the line there, so the centre of the data is fine.

Step 2: Look at the two ends. Both ends are above the line, which draws a curve like a smile (curving up). A smile-shaped curve means that the upper values are stretched out much further than a normal curve would allow and the lower values are packed in closer.

Step 3: A stretched upper end means a long tail on the right side. This is typical of data such as mobile data use, where most people use a moderate amount and a few heavy users use a lot.

Step 4: Because the right end is much higher above the line than the left end is, the data are right-skewed, not heavy-tailed. For heavy tails the left end would have to fall below the line.

Result: The data are not normal. The shape is right-skewed, with a few very heavy users forming the long right tail.

Chart 2: Plant Height (cm), n = 30

Aim: To decide whether the heights of 30 plants are normal, and to name the shape.

What I See in the Plot

(a) The dots in the middle (about 35 to 38 cm) follow the line well.

(b) At the left end the dots fall below the line. The lowest plant (about 17 cm) is very far below it and the next one (about 30 cm) is also below.

(c) At the right end the dots rise above the line. The tallest plants (about 49, 50 and 52 cm) are well above it.

Answer (as asked in the sheet)

Normal? (Yes / No)	**No**
Shape (from the list)	Heavy-tailed
Evidence in the plot	The dots make an S-shape: the left end falls below the line (about 17 cm and 30 cm) and the right end rises above it (about 49, 50 and 52 cm), while the middle bunch (35 to 38 cm) sits on the line.

Explanation

Step 1: Check the middle: the dots follow the line, so most plants have ordinary heights packed closely around 36 cm.

Step 2: Check the left end: the dots go below the line, so the smallest plants are smaller than a normal curve predicts.

Step 3: Check the right end: the dots go above the line, so the tallest plants are taller than a normal curve predicts. Both ends leave the line in opposite directions, which draws an S-shape.

Step 4: An S-shape means the tails are heavier than normal: there are more extreme values (both very short and very tall plants) than a normal distribution would give.

Step 5: Why not "normal with outlier(s)"? That answer fits only when one or two dots leave an otherwise straight line. Here several dots on each end (at least five) depart, and they depart on both sides, so it is the whole tail behaviour that is not normal. Why not skewed? A skewed set would bend the same way (both ends above or both below the line), but here the ends go in opposite directions.

Result: The data are not normal. The shape is heavy-tailed, because both tails stretch away from the line in opposite directions.

Chart 3: Typing Speed (Words per Minute), n = 30

Aim: To decide whether the typing speeds of 30 people are normal, and to name the shape.

What I See in the Plot

(a) Nearly all dots lie right along the red line, from the lowest (about 34 wpm) to the highest (about 78 wpm).

(b) The only small differences are a few dots a little above the line at the left end (about 34, 40 and 43 wpm) and some small wiggles in the middle. They are tiny and do not bend in one direction.

(c) There is no sharp curve, no S-shape, no gap and no dot far away from the others.

Answer (as asked in the sheet)

Normal? (Yes / No)	**Yes**
Shape (from the list)	Normal
Evidence in the plot	The dots follow the straight reference line from end to end, with only small random wiggles, no curve at either end and no far-away dot.

Explanation

Step 1: Look at the middle: the dots sit on the line.

Step 2: Look at the two ends: the dots stay close to the line there as well. The small gaps at the left end are the size of normal sampling variation for a sample of only 30 values; real normal data never lie perfectly on the line.

Step 3: No bend, no S-shape and no outlier means that nothing in the plot disagrees with a normal distribution, so the correct shape from the list is Normal.

Step 4: Because the data are normal, the mean and the standard deviation can be used to describe them, and normal-theory methods can be used.

Result: The data are normal. The Q-Q plot is a straight line with only small random wiggles.

Chart 4: Resting Pulse (Beats per Minute), n = 100

Aim: To decide whether the resting pulse of 100 people is normal, and to name the shape.

What I See in the Plot

(a) In the middle (about 58 to 66 bpm) the dots lie close to the line.

(b) At the left end the dots drop below the line. The lowest values (about 44 to 52 bpm) are well below where the line (about 53 bpm at the far left) says they should be.

(c) At the right end the dots go flat at about 68 to 69 bpm and stay far below the line, which keeps on rising to about 76 bpm. Many dots are piled up near 69 bpm.

Answer (as asked in the sheet)

Normal? (Yes / No)	**No**
Shape (from the list)	Left-skewed
Evidence in the plot	The dots curve downward below the line at both ends: the left end falls to about 44 bpm, well under the line, and the right end flattens out at about 69 bpm while the line keeps rising. This makes a frown-like curve, with a long tail on the left.

Explanation

Step 1: The middle of the plot follows the line, so most people have a pulse around 62 to 66 bpm.

Step 2: At the right end the dots stop rising (they pile up near 68 to 69 bpm), so there are no large values. The upper end of the data is cut short, like a ceiling.

Step 3: At the left end the dots fall far below the line, so the data have a long stretch of low values (down to about 44 bpm).

Step 4: Both ends are below the line, which draws a frown (curving down). A frown means a short upper tail and a long lower tail, which is a left-skewed distribution.

Step 5: Why not light-tailed? A light-tailed set would have its left end above the line and its right end below it. Here the left end goes below the line, so the lower tail is long, not short.

Result: The data are not normal. The shape is left-skewed, with a long tail of low pulse values and a ceiling near 69 bpm.

Part B (4 Marks)

B1: Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

Aim: To name the way Chart 1 departs from normality and to choose a suitable remedy.

Answer

Departure: Chart 1 is right-skewed. The dots bend above the line at the upper end, and two very heavy users (about 1550 and 1700 MB) lie far above it. **Remedy: apply a log transform to the values (log of the MB used), then draw the Q-Q plot again.**

Result: Chart 1 is right-skewed. The first remedy to try is a log transform of the data use, followed by a new Q-Q plot; if that is not enough, use the median and IQR (robust methods) or split the data.

Explanation

Step 1: Right-skewed data have a long tail of large values. Taking the logarithm shrinks large values much more than small ones: for example, 1000 MB becomes log(1000) = 6.91, and 1700 MB only becomes log(1700) = 7.44, which is a gap of 0.53 instead of 700 MB. So the long right tail is pulled in.

Step 2: The log transform is the first thing to try for right-skewed, positive data like data use, because it is simple, works on every positive value, and can be undone with exp().

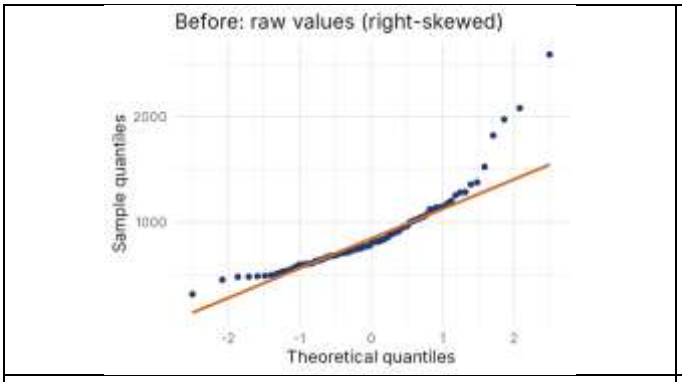
Step 3: After the transformation I would draw the Q-Q plot again. If the dots now follow the line, the mean and standard deviation of log(value) can be used safely.

Step 4: If the log does not fix it (for example the two extreme users are real outliers), the second choice is a robust method: describe the data with the median and the IQR instead of the mean and the SD, or analyse the heavy users as a separate group (split the data).

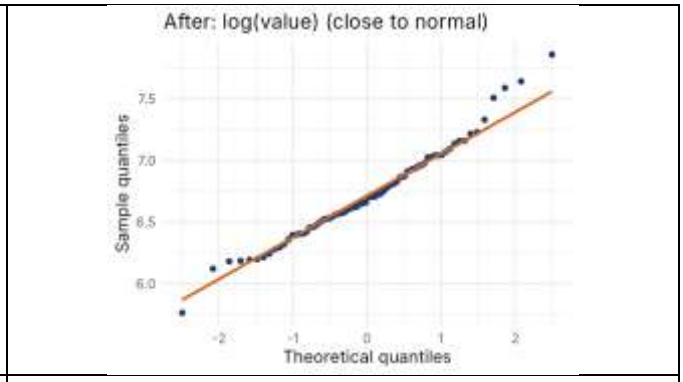
Step 5: A square transform is not suitable here: squaring stretches large values even more, so it is used for left-skewed data, not right-skewed data.

Illustration of the Remedy

The two Q-Q plots below were made in R from 80 simulated right-skewed values, just to show how the log transform works. They are not the data of Chart 1, whose original values are not given. On the left, the raw values bend above the line at the upper end like Chart 1. On the right, after log(), the dots follow the line.



Before: raw values (right-skewed)



After: log(value) (nearly normal)

B2: Fill in the Blanks for the Q-Q Plot Code

Question: Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Aim: To complete the ggplot2 code that draws a normal Q-Q plot together with its reference line.

Answer

```
ggplot(df, aes(sample = value)) + stat_qq() + stat_qq_line()
```

The first blank is **stat_qq()** and the second blank is **stat_qq_line()**.

Explanation

Step 1: aes(sample = value) tells ggplot2 that the column value is the sample to be compared with the normal distribution. A Q-Q plot uses the aesthetic sample, not x or y.

Step 2: stat_qq() draws the dots: it sorts the data and plots every sorted value (sample quantile) against the matching quantile of a standard normal distribution (theoretical quantile).

Step 3: stat_qq_line() draws the red "perfect normal fit" line. It passes through the first and third quartiles of the data, so the dots can be compared with a straight reference.

Step 4: Both layers are added with +, in the same way as any other ggplot2 layer. (geom_qq() and geom_qq_line() are the same layers under their geom names, so those also give full marks.)

Complete Program

```
library(ggplot2)

ggplot(df, aes(sample = value)) +
  stat_qq(colour = "#1F3A6E") +                    # the dots
  stat_qq_line(colour = "#D2691E", linewidth = 1) +   # the reference line
  labs(x = "Theoretical quantiles", y = "Sample quantiles") +
  theme_minimal()
```

Result: The blanks are stat_qq() and stat_qq_line(). Together they draw the dots and the reference line of a normal Q-Q plot.

Summary of Answers

Chart	Normal?	Shape	Main evidence
1: Mobile data (n = 80)	No	Right-skewed	Both ends above the line; two huge values (about 1550, 1700 MB)
2: Plant height (n = 30)	No	Heavy-tailed	S-shape: left end below, right end above the line
3: Typing speed (n = 30)	Yes	Normal	Dots follow the line; only small wiggles
4: Resting pulse (n = 100)	No	Left-skewed	Both ends below the line; right end flat at about 69 bpm

Part B: B1 - right-skewed, try a log transform (then median/IQR or splitting the data if needed). B2 - stat_qq() and stat_qq_line().

Conclusion

A normal Q-Q plot is read by watching how the dots leave the red line. Dots on the line mean normal data (Chart 3). Both ends above the line mean a long right tail (Chart 1), both ends below the line mean a long left tail (Chart 4), and an S-shape means heavier tails than normal (Chart 2). Knowing the shape also tells what to do next: a right-skewed variable like mobile data use is best handled with a log transform before using mean-and-SD methods.